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Ethiopian Multilingual Translation System

Using Transformer-Based Models and Ensemble Translation

A Project Report Submitted to the Department of Data Science

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Debre Birhan , Ethiopia

DECLARATION

We hereby declare that this research project entitled *"Ethiopian Multilingual Translation System Using Transformer-Based Ensemble Models"* is our original work and has been carried out under the guidance and supervision of our project advisor. To the best of our knowledge and belief, this work has not been submitted, either in whole or in part, for the award of any degree, diploma, or other academic qualification at this or any other institution.

All sources of information, references, and materials used in the preparation of this project have been properly acknowledged and cited.

Department: Data Science
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ABSTRACT

Ethiopia is a linguistically diverse nation with over 80 languages spoken across its regions, yet the majority of these languages remain severely underserved by existing digital language technologies. The absence of high-quality, automated translation tools for Ethiopian languages continues to hinder communication, access to information, and digital inclusion for millions of citizens. This project addresses that gap through the design, development, and evaluation of an Ethiopian Multilingual Translation System capable of translating text among five major Ethiopian languages—Amharic, Afaan Oromo, Tigrinya, Somali, and Afar—and English.

The system leverages three state-of-the-art transformer-based neural machine translation models: NLLB-200 (No Language Left Behind), M2M100, and T5 (Text-to-Text Transfer Transformer). Rather than relying on a single model, an ensemble translation strategy is employed to combine the outputs of all three models, selecting or synthesising the most accurate translation using confidence-based and similarity-based decision mechanisms. This approach capitalises on the complementary strengths of each model, thereby improving overall translation quality, fluency, and contextual accuracy.

The study encompasses the complete system development pipeline, including multilingual parallel dataset collection, data cleaning and normalisation, language pair generation, model integration, automatic language detection, and user interface development using the Gradio framework. System performance is rigorously evaluated using two standard automatic metrics: BLEU (Bilingual Evaluation Understudy) and ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation), applied across multiple Ethiopian language pairs on a held-out testing dataset.

Experimental results consistently demonstrate that the ensemble model outperforms each individual translation model across most language pairs, achieving higher BLEU and ROUGE-L scores. These findings confirm that integrating multiple pre-trained multilingual models through an ensemble framework is an effective strategy for improving machine translation quality in low-resource language settings. The developed system provides a scalable, practical, and accessible multilingual translation solution that contributes to advancing Natural Language Processing research and digital communication for Ethiopian languages.

Keywords: Machine Translation, Neural Machine Translation, Transformer, NLLB-200, M2M100, T5, Ensemble Learning, Ethiopian Languages, Low-Resource NLP, BLEU, ROUGE-L, Gradio, Multilingual Systems.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Language plays a vital role in communication, education, business, and access to information. Ethiopia is a multilingual country with numerous widely spoken languages, including Amharic, Afaan Oromo, Tigrinya, Somali, Afar, and English. Despite the linguistic diversity of the country, digital language resources and translation technologies for many Ethiopian languages remain limited compared to those available for high-resource languages such as English, French, and Chinese. Due to the rise of its applications in many fields, Natural Language Processing (NLP), a sub-field of Artificial Intelligence (AI), is receiving a lot of attention in terms of research and development (Kalyanathaya et al., 2019). NLP tasks such as Machine Translation (MT), word frequency analysis, Parts of Speech (POS) Tagging, Chunking, Named Entity Recognition (NER), Emotion Detection, and Semantic Role Labeling is currently highly researched areas in different high-resource languages.

Recent advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have significantly improved machine translation through the use of Neural Machine Translation (NMT) models. Modern transformer-based models such as google/mt5, NLLB-600 (No Language Left Behind), M2M100, and T5 have demonstrated remarkable performance in multilingual translation tasks by enabling direct translation between multiple languages without relying on English as an intermediate language.

However, many Ethiopian languages continue to face challenges due to limited parallel corpora, insufficient language resources, and inadequate translation systems. Existing translation tools often provide inconsistent results, limited language coverage, or reduced translation quality for low-resource languages. This creates barriers to information sharing, education, digital inclusion, and cross-cultural communication within Ethiopia.

To address these challenges, this project focuses on the development of an Ethiopian Multilingual Translation System capable of translating text among multiple Ethiopian languages and English. The system utilizes advanced transformer-based models, including NLLB-200, M2M100, and T5, and combines their strengths through an ensemble translation approach. The project also involves data collection, preprocessing, multilingual dataset construction, language pair generation, performance evaluation using BLEU and ROUGE-L metrics, and the development of an interactive user interface using Gradio.

In this paper, we overview research works done in the area of selected NLP tasks for four Ethiopian languages. We cover mainly the following 6 lan guages, namely Amharic

(Amh), Afaan Oromo (Orm), Tigrinya (Tir), and Afar(aar), Somali(som) and the other one is English(eng) as a base language. of this paper are as follows: (1) Reporting the cur rent state-of-the-art NLP research for Ethiopian languages. (2) Discussing NLP progress in Ethiopian languages and the main challenges and opportunities for Ethiopian NLP research. (3) Collecting and presenting publicly available resources for different NLP tasks in Ethiopian languages in one GitHub repository that can be extended periodically in collaboration with other researchers.

The proposed system aims to improve translation accessibility and quality for Ethiopian languages, promote digital inclusiveness, and contribute to the advancement of Natural Language Processing research for low-resource languages in Ethiopia.

1.2 Statement of the Problem

Ethiopia is home to many languages spoken by millions of people across different regions. Effective communication among speakers of different languages is essential for education, business, government services, healthcare, and access to digital information. However, most Ethiopian languages are considered low-resource languages, meaning that they lack sufficient digital datasets, language technologies, and high-quality machine translation systems.

Although machine translation has advanced significantly through the development of Neural Machine Translation (NMT) and transformer-based models, the majority of existing translation systems primarily focus on high-resource languages. As a result, translation quality for Ethiopian languages such as Amharic, Afaan Oromo, Tigrinya, Somali, and Afar remains relatively low. Many available translation tools provide limited language support, inaccurate translations, or require translation through an intermediate language, which can introduce errors and reduce translation quality.

Furthermore, the scarcity of parallel multilingual datasets for Ethiopian languages creates challenges in developing robust translation models capable of supporting direct language-to-language translation. This limitation affects communication, information accessibility, and digital inclusion for millions of users who rely on these languages.

Therefore, there is a need to develop an efficient multilingual translation system that can support direct translation among Ethiopian languages and English while improving translation accuracy and accessibility. By leveraging advanced transformer-based models such as NLLB-600, M2M100, and T5, this study seeks to address the challenges of low-resource language translation and contribute to the development of effective language technologies for Ethiopia.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop an Ethiopian Multilingual Translation, language identification and text summarizing System that provides accurate and efficient translation among selected five Ethiopian languages and English using advanced transformer-based machine translation models.

1.3.2 Specific Objectives

- To collect and prepare multilingual datasets for Ethiopian languages and English from available translation resources.
- To preprocess and clean the collected datasets by removing duplicates, handling missing values, and normalizing text data and other relevant preprocessing.
- To generate multilingual language pairs that support bidirectional translation among the selected languages.
- To implement and integrate transformer-based translation models, including NLLB-200, M2M100, and T5, for multilingual translation tasks.
- To develop an ensemble translation framework that combines the strengths of multiple translation models to improve translation quality.
- To implement an automatic language detection mechanism for identifying the source language of input text.
- To design and develop a user-friendly web interface for easy interaction with the translation system.
- To contribute to the advancement of Natural Language Processing technologies for low-resource Ethiopian languages.

1.4 Research Questions

To address the challenges of multilingual communication ,identification and translation among Ethiopian languages, this study seeks to answer the following key research questions:

- How can an effective multilingual translation system be developed to support translation among selected Ethiopian languages and English?
- What data preprocessing and multilingual dataset preparation techniques are necessary to build a reliable translation system?
- How accurately can the proposed system perform translations between different Ethiopian language pairs, including Amharic, Afaan Oromo, Tigrinya, Somali, Afar, and English?
- How effective is the automatic language detection mechanism in identifying the source language before translation?

- Compaire letters, words and sentence length b/n languages and see which language use more words?

These research questions guide the design, implementation, evaluation, and analysis of the Ethiopian Multilingual Translation System developed in this study. They focus on assessing both the technical performance of the translation models and the practical usefulness of the system in supporting communication among speakers of different Ethiopian languages.

1.5 Scope of the Study

This study focuses on the design, development, and evaluation of an Ethiopian Multilingual Translation System capable of translating text among selected Ethiopian languages and English. The system primarily supports Amharic, Afaan Oromo, Tigrinya, Somali, Afar, and English, which are among the most widely used languages in Ethiopia. The project utilizes multilingual datasets collected from available translation resources and applies preprocessing techniques such as data cleaning, normalization, duplicate removal, and language pair generation to prepare the data for machine translation tasks, word frequency analysis, NER assesment and so on.

The study implements and integrates advanced transformer-based machine translation models, including NLLB-200, M2M100, and T5, to perform multilingual text translation. An ensemble translation approach is employed to combine the strengths of multiple models and improve translation quality. The system also incorporates an automatic language detection mechanism to identify the source language before translation and a web-based user interface developed using Gradio to facilitate user interaction.

The performance of the proposed system is evaluated using standard machine translation metrics, including BLEU and ROUGE-L scores and chrF, as well as qualitative analysis of translation outputs. The study is limited to text-based translation and does not include speech recognition, speech-to-text translation, text-to-speech synthesis, image translation, or real-time conversational translation. Furthermore, the translation quality depends on the availability and quality of multilingual datasets for the selected Ethiopian languages. Despite these limitations, the study aims to provide an effective multilingual translation solution, NER, WWord frequency that contributes to improving digital communication and language technology development for Ethiopian languages.

1.6 Significance of the Study

The development of an Ethiopian Multilingual Translation System is significant because it addresses the communication barriers that exist among speakers of different Ethiopian languages. Ethiopia is a linguistically diverse country with numerous languages spoken across various regions, making effective communication and access to information challenging for many

individuals. By providing an automated translation system that supports multiple Ethiopian languages and English, this study contributes to enhancing communication, information sharing, and digital inclusion.

The proposed system benefits individuals, educational institutions, government organizations, businesses, and researchers by enabling easier access to multilingual information. Students and educators can use the system to access learning materials available in different languages, while government institutions can improve communication and service delivery to citizens from diverse linguistic backgrounds. Businesses can also benefit from improved communication with customers and partners who speak different languages.

From a technological perspective, the study contributes to the advancement of Natural Language Processing (NLP) and machine translation research for low-resource Ethiopian languages. By utilizing advanced transformer-based models such as NLLB-600, M2M100, and T5, the project demonstrates the application of modern artificial intelligence techniques to address language-related challenges.

Additionally, the study promotes digital accessibility by helping speakers of Ethiopian languages access online content, educational resources, and digital services that may otherwise be unavailable due to language barriers. The findings and outcomes of this research can serve as a foundation for future developments in multilingual language technologies, including speech translation, conversational AI systems, and other language-based applications. Ultimately, the study contributes to preserving linguistic diversity while supporting technological innovation and inclusive access to information in Ethiopia.

1.7 Limitations of the Study

Despite the effectiveness of the proposed multilingual translation system, the study faced several limitations that may affect its overall performance and generalization.

Firstly, the availability of high-quality parallel corpora for Ethiopian languages was limited. Many language pairs lacked sufficient training data, which reduced translation accuracy for low-resource languages and affected model robustness.

Secondly, computational resources constrained the training and evaluation process. Due to hardware limitations, especially GPU memory and processing power, large-scale fine-tuning of models such as NLLB-600, google/mts-small for translation, M2M100 and others was restricted, leading to reliance on pre-trained models rather than fully optimized custom training.

Thirdly, linguistic complexity and diversity of Ethiopian languages posed a challenge. Variations in dialects, script differences (such as Ge'ez-based scripts), and informal text structures made preprocessing and normalization more difficult and sometimes inconsistent.

Fourtly, We(investigators) are challanged by the contextual meaning, defination and grammer usabelity.

Additionally, evaluation metrics such as BLEU and ROUGE-L, chrf, while widely used, do not fully capture semantic meaning or contextual accuracy in translation quality. This may result in incomplete assessment of model performance.

Finally, the system was primarily tested in a controlled environment using a limited dataset. Real-world deployment scenarios, such as noisy user input, code-switching, and informal communication, were not extensively evaluated.

These limitations suggest that further improvements in data quality, computational resources, and evaluation techniques are necessary for enhancing the system's overall performance.

1.8 Organization of the Document

We see the thesis of This document is organized into seven main chapters, each focusing on a specific part of the study to ensure a clear and logical presentation of the research work.

Chapter One introduces the study by providing background information, stating the problem, defining objectives, research questions, scope, significance, limitations, and the overall structure of the document.

Chapter Two presents a detailed review of related literature, including concepts such as Natural Language Processing, Machine Translation, Neural Machine Translation, transformer-based models, multilingual translation systems, Ethiopian language research, and evaluation metrics. It also identifies the research gap and presents the conceptual framework.

Chapter Three describes the methodology used in the study. It includes the research design, system development approach, data collection methods, dataset description, preprocessing steps, language pair generation, dataset splitting, system requirements, and development tools used.

Chapter Four focuses on system analysis and implementation process. It discusses the proposed system, functional, system architecture, data flow, translation workflow and user interface design. the implementation process covers environment setup, exploratory data analysis, model implementation (NLLB-200, M2M100, and T5), , language detection, translation engine development, Gradio interface.

Chapter five presents the results and discussion of the study. It includes language distribution analysis, evaluation of translation performance, BLEU and ROUGE-L scores, model comparison, translation examples, system performance analysis, and discussion of findings.

Chapter six concludes the study and provides recommendations. It summarizes the research, highlights contributions and suggests future improvements such as expanding to additional

Ethiopian languages, integrating speech-to-text translation, developing mobile applications, and deploying the system on cloud platforms.

Finally, the document includes References and Appendices, which provide supporting materials such as source code, dataset samples, language mappings, evaluation results, user manual, installation guide, and hardware specifications.

CHAPTER 2: LITERATURE REVIEW

This chapter presents the theoretical foundations and related studies relevant to multilingual machine translation. It reviews key concepts such as machine translation, neural machine translation, transformer architecture, multilingual translation systems, and ensemble-based approaches. It also examines existing research on Ethiopian language translation, identifies key research gaps, and summarizes findings that motivate the development of the proposed system.

2.1 Theoretical Framework

2.1.1 Machine Translation

Machine Translation (MT) is a subfield of Natural Language Processing (NLP) that focuses on automatically translating text from one language to another. Early approaches were based on Rule-Based Machine Translation (RBMT), which relied on manually designed linguistic rules and bilingual dictionaries. These systems were highly dependent on linguistic expertise and struggled with scalability.

Subsequently, Statistical Machine Translation (SMT) emerged, which uses probabilistic models learned from large parallel corpora. Although SMT improved translation flexibility compared to RBMT, it still faced limitations in fluency and contextual understanding.

Modern machine translation systems are primarily based on Neural Machine Translation (NMT), which leverages deep learning techniques to learn end-to-end mappings between source and target languages, producing more fluent and context-aware translations.

2.1.2 Neural Machine Translation

Neural Machine Translation (NMT) is an advanced approach that utilizes deep neural networks to model the translation process directly. Unlike traditional methods, NMT processes entire sentences as input, enabling the model to capture long-range dependencies and contextual relationships between words.

A major advancement in NMT is the introduction of attention mechanisms, which allow the model to focus on relevant parts of the input sentence during translation. This significantly improves translation accuracy, especially for long and complex sentences.

As a result, NMT has become the dominant approach in modern machine translation systems due to its superior fluency and contextual understanding.

2.1.3 Transformer Architecture

The Transformer architecture forms the foundation of most state-of-the-art machine translation systems. Unlike recurrent neural networks, Transformers rely entirely on self-attention mechanisms to model relationships between words in a sequence, regardless of their positions.

This architecture enables parallel computation, which significantly improves training efficiency and scalability. Additionally, self-attention allows the model to capture both local and global dependencies effectively.

Transformer-based models such as NLLB-200, M2M100, and T5 have achieved strong performance in multilingual machine translation tasks and are widely adopted in both research and practical applications.

2.1.4 Multilingual Translation Systems

Multilingual translation systems are designed to handle multiple language pairs within a single unified model. These systems learn shared representations across languages, enabling knowledge transfer from high-resource languages to low-resource languages.

This approach improves scalability and reduces the need to train separate models for each language pair. It is particularly beneficial for low-resource language settings, such as Ethiopian languages, where training data is limited.

By leveraging cross-lingual learning, multilingual models can significantly improve translation quality for under-resourced languages.

2.1.5 Ethiopian Language Translation Research

Research on Ethiopian language translation has increased in recent years due to the growing demand for digital communication and localization in languages such as Amharic, Afaan Oromo, and Tigrinya. However, despite this progress, the field remains underdeveloped compared to high-resource languages.

One of the major challenges is the scarcity of large, high-quality parallel corpora. Most available datasets are small, domain-specific, or noisy, which limits the performance of both statistical and neural machine translation systems.

Another significant challenge is the linguistic complexity of Ethiopian languages. These languages exhibit rich morphology, diverse syntactic structures, and variations in spelling and dialects. Amharic, for example, uses the Ge'ez script, which requires specialized preprocessing compared to Latin-based languages.

Early research efforts mainly focused on rule-based and statistical approaches, but these methods showed limited generalization capability due to data scarcity and linguistic complexity.

More recent studies have shifted toward neural and transformer-based models, which have shown improved performance.

Despite these advancements, existing systems still face challenges in achieving high accuracy and robustness. This study builds upon these developments by exploring an integrated multilingual approach using advanced transformer-based models and ensemble techniques.

2.2 Review of Related Work

Recent research in machine translation has increasingly focused on transformer-based multilingual models due to their strong performance in low-resource settings. These models have been widely applied to improve translation quality across diverse language pairs.

Solomon Teferra Abate et al. (2018) developed parallel corpora for English and several Ethiopian languages, including Amharic, Tigrinya, Afaan Oromo, Wolaytta, and Ge'ez, to support bidirectional statistical machine translation. The study found that the availability of well-aligned parallel corpora significantly improves translation performance. However, translation into Ethiopian languages achieved lower accuracy due to their rich morphology and complex linguistic structures. The researchers concluded that larger and higher-quality corpora are essential for improving machine translation systems for low-resource Ethiopian languages. While the study focused mainly on bilingual English–Ethiopian language pairs, the present research extends this work by constructing a multilingual corpus for six languages and supporting 30 bidirectional translation directions in a many-to-many translation framework.

Atnafu Lambebo Tonja (2024) proposed EthioMT, a parallel corpus aimed at improving machine translation for low-resource Ethiopian languages by providing a unified dataset for 15 languages and benchmark data for 23 languages. The main goal was to address the lack of publicly available and standardized datasets for Ethiopian language translation research. They used Transformer models trained from scratch and a fine-tuned multilingual model (M2M100) to evaluate translation performance in both EnglishEthiopian language directions. The findings showed that fine-tuned M2M100 significantly outperformed the Transformer baseline, and translation into English performed better than translation into Ethiopian languages, especially for higher-resource languages. However, the study is limited by data imbalance across languages, small corpus sizes for several languages, and domain restrictions (mostly religious and mixed text), which affect overall model performance and generalization.

Matiyas Gutema Angecha (2025) developed a Transformer-based neural machine translation system for Amharic and Tigrinya. Their findings demonstrated that Transformer architectures significantly outperform traditional statistical machine translation approaches, particularly in handling morphologically complex and low-resource languages. In addition, the study confirmed that data augmentation techniques, such as back-translation, effectively mitigate data scarcity and substantially improve translation quality. However, the study is limited to only two closely

related languages (Amharic and Tigrinya) and relies on a relatively small, domain-specific parallel corpus, which restricts its generalization to broader multilingual settings. Furthermore, the model does not address a wider multilingual translation scenario or incorporate cross-lingual transfer across diverse language families. These limitations motivate the present research, which extends translation capability to a six-language multilingual framework to improve scalability and real-world applicability.

2.3 Research Gap

Although Abate et al. (2018) developed one of the earliest and most important parallel corpora for Ethiopian languages and demonstrated improvements in statistical machine translation, several limitations remain. First, the study primarily relies on statistical machine translation (SMT), which is less effective in capturing long-range dependencies and complex linguistic structures compared to modern neural approaches. Second, the corpus size is still relatively limited for training high-quality translation models, especially for low-resource Ethiopian languages. Third, the work focuses mainly on bilingual English–Ethiopian language pairs, which restricts its applicability to multilingual or many-to-many translation scenarios.

Although EthioMT (Tonja, 2024) provides a significant contribution by introducing a large-scale multilingual corpus and benchmarking multiple Ethiopian languages, several limitations remain. The dataset suffers from severe imbalance across languages, where some languages have significantly fewer resources than others. In addition, many language pairs still rely on small or domain-specific corpora (mainly religious and mixed text domains), which limits model generalization to real-world applications. Furthermore, although the study evaluates multilingual models such as M2M100, it does not deeply explore task-specific optimization strategies such as advanced tokenization, morphology-aware modeling, or domain adaptation techniques.

Angecha and Tachbelie (2025) demonstrated that Transformer-based models significantly outperform statistical approaches for Amharic–Tigrinya translation, especially when combined with subword tokenization and data augmentation. However, the study is limited in several aspects. First, it focuses only on two closely related languages, which restricts the generalizability of the model to broader multilingual environments. Second, the dataset used is relatively small and domain-specific, reducing robustness across different text domains. Third, the system does not incorporate cross-lingual transfer learning or multilingual training strategies, which are crucial for improving performance in low-resource settings. Finally, the model does not address scalability to many-to-many translation systems, which is essential for real-world multilingual applications.

CHAPTER 3: METHODOLOGY

This chapter describes the methodology used in developing the multilingual machine translation system. It explains the research design, data collection process, dataset preparation, preprocessing techniques, model integration, and evaluation methods. The goal of this methodology is to provide a structured approach for building and testing a robust translation system capable of handling Ethiopian and other multilingual language pairs.

3.1 Research Design

This study adopts a developmental and experimental research design aimed at building and evaluating a multilingual machine translation system for Ethiopian languages. The design focuses on both system development and performance assessment, ensuring that the proposed solution is not only implemented but also rigorously tested using standard evaluation methods.

The developmental aspect involves designing and implementing a software-based translation system that integrates multiple transformer-based models,The system development methodology used in this study is based on a modular and incremental development approach. In the modular approach, the entire system is divided into independent components, each responsible for a specific function. These components include data preprocessing, language detection, model integration (NLLB-200, M2M100, and T5), ensemble translation processing, evaluation metrics computation, and user interface development. Each module is developed, tested, and validated separately before being integrated into the full system. and later integrated into a unified framework.

The experimental aspect focuses on evaluating the performance of the system under different conditions. This includes testing translation outputs across multiple language pairs and comparing the results of individual models versus the ensemble approach. The evaluation is conducted using quantitative metrics such as BLEU and ROUGE-L scores to measure accuracy, fluency, and similarity to reference translations.

3.2 Data Sources and descrption

The dataset used in this study is a multilingual parallel news corpus obtained from Kaggle. It contains aligned sentence pairs across five Ethiopian languages: Amharic (AMH), Afaan Oromo (ORM), Somali (SOM), Tigrinya (TIR), Afar (AAR), and English (ENG). This is designed to support machine translation and multilingual natural language processing research.Totally, there are 12,186 sentences as a whole and Each language file contains 2,031 parallel sentences resulting in a balanced dataset across all language pairs. The texts primarily consist of news

articles covering international events, politics, culture, science, archaeology, and environmental issues. Every sentence in one language corresponds semantically to the same sentence in the other languages, enabling accurate parallel translation training.

3.3 Data Preprocessing

Data preprocessing is an essential stage in the development of the multilingual machine translation system. It involves transforming raw textual data into a clean, consistent, and structured format suitable for training and evaluating machine learning models. Proper preprocessing improves model performance, reduces noise, and ensures higher translation accuracy.

3.3.1 Missing Value Handling

Data cleaning is performed to remove irrelevant and noisy elements from the dataset. The goal is to ensure that the dataset contains only meaningful text that contributes to translation learning. This means that 2030 sentence pairs are valid and aligned correctly, while 1 record is invalid. This indicates that one row contains missing or empty text across all six languages and row has removed before training translation model,then 2030 cleaned sentences are obtained in each languages.

3.3.2 Duplicate Removal

Duplicate entries in the dataset are detected and removed to avoid redundancy. Repeated sentence pairs can introduce bias during training, causing the model to overfit certain patterns. After removing missing, Duplicate row analysis revealed and no duplicate records in the dataset, indicating that all 2,030 valid sentence pairs are unique and suitable for model training.

3.3.3 Text Normalization

Text normalization is applied to standardize the dataset into a consistent format. This includes converting text into uniform Unicode normalization (NFC), handling punctuation, and normalizing spacing , converting all text to lowercase was performed while preserving language-specific characters and linguistic information. . This step ensures that similar sentences are treated uniformly by the model, improving translation quality and reducing ambiguity.

3.3.4 data frame creation

A Pandas DataFrame is created to store multilingual data in a structured format.the orginal dataset is in separetd text file, and after dataframe created The dataset included six language columns: Amharic, English, Oromo, Somali, Tigrinya, and Afar . Each row has 1230 news sentences that represents aligned sentences with the same meaning across different languages. This alignment supports multilingual NLP and machine translation tasks.

3.3.5 data set analysis

The analysis presents a comprehensive statistical overview of multilingual text datasets across six languages. The study focuses on special character distribution, word frequency, vocabulary size, and language-specific lexical patterns, which are essential for preprocessing in natural language processing (NLP) and machine translation systems.

First, the special character analysis shows that each language has distinct punctuation usage patterns. Amharic and Tigrinya contain Ethiopic punctuation marks such as “-”, “/”, and “?”, while English, Oromo, Somali, and Afar rely heavily on Latin-based punctuation like periods, commas, quotation marks, and hyphens. English and Oromo show high use of standard punctuation such as periods and commas, whereas Amharic and Tigrinya heavily use script-specific symbols, reflecting their unique writing systems. Across all languages, symbols like quotes, parentheses, percentages, and currency signs appear consistently, indicating shared formatting conventions in multilingual corpora.

Second, the symbol frequency and uniqueness analysis reveals differences in dataset complexity. each language have their on unique symbols.

Third, the word frequency analysis highlights dominant linguistic patterns in each language. In Amharic, common words such as “”, “”, and “” reflect relational and functional usage. English is dominated by function words like “the”, “of”, “to”, and “in”, which is typical in natural language corpora. Oromo shows frequent use of words like “fi”, “akka"and “kan”, while Somali is dominated by tokens of function similar to syllables such as “ee”, “oo" and “ayaa”. Tigrinya shows high frequency of words like “”, “”, and “”, and Afar includes frequent terms like “kee”, “tan”, and “yan”. These patterns indicate that all datasets are linguistically rich and properly aligned for NLP training.

Fourth, the vocabulary size comparison demonstrates dataset diversity or word unigueness. Afar has the largest vocabulary (16,780 unique words), followed by Oromo (14,338), Amharic (13,108), Tigrinya (12,272), Somali (11,107), and English (8,829). This suggests that Afar and Oromo datasets may contain more morphological variation or less normalization compared to English.

fifth, summary statistics table summary data For every language, it calculates multiple metrics:Total rows: Number of sentence pairs or entries in the dataset 2030 per language. Total words: Number of tokens after splitting text. Unique words: Vocabulary size /number of distinct tokens. Total letters: Character count excluding spaces, representing raw textual length. Special characters: Count of non-alphanumeric symbols using regex filtering (A-Za-z0-9]). Combined metric all word and char): Sum of words and letters to estimate dataset size. Top word: The most frequent token in each language. Top frequency: Occurrence count of the most common word. From the results, we observe that:

language	total_rows	total_words	unique_words	total_letters	special chars	all word$char	top_word	top_frequency
amh	2030	39102	15069	163596	160623	202698	አና	809
eng	2030	47627	12335	254943	6110	302570	the	2946
orm	2030	42018	15274	308328	6788	350346	fi	994
som	2030	55006	14563	307138	5677	362144	ee	1976
tir	2030	43511	14235	166825	163844	210336	ኣብ	2056
aar	2030	38381	18422	245102	4199	283483	kee	970
TOTAL	12180	265645	89898	1445932	347241	1711577	-	9751

Figure 3.1: data set analysis

Finally, After computing per-language statistics, values across all languages, giving a global view of dataset size. This includes total words (265,645), total unique words (89,898), total letters (1,445,932), and total special characters (347,241).

3.4 Language Pair Generation

Language pair generation is the process of structuring the dataset into consistent pair. This process involves identifying and defining all possible translation directions between the selected languages, including Amharic, Afaan Oromo, Tigrinya,afar,somali and English and this allowing the system to support many-to-many translation.Language pair which is large-scale multilingual parallel dataset by generating bidirectional translation pairs across six languages: Amharic (amh), English (eng), Oromo (orm), Somali (som), Tigrinya (tir), and Afar (aar) . The main objective is to expand a single aligned multilingual corpus into multiple translation directions, which is essential for training robust multilingual machine translation models.

After data collection, cleaning, and preprocessing, the final dataset consists of aligned sentence pairs covering multiple Ethiopian languages and English. The dataset is organized into structured columns, including language_pair, source_text, and target_text and 60900 total sentences. which ensure consistency and compatibility with machine translation models. Each row represents a single translation example used for training, validation, or testing purposes.

FULL DATASET

==========

	language_pair	source_text	target_text
0	amh-eng	የእስራኤል አየር ሃይል የተቃጣበትን የሚሳኤል እና የሞርታር ጥቃት ተከትሎ ነው ቤል በጋዛ ሰርጥ ላይ ጥቃት መፈፀም ጀመረ።	The Israeli Air Force launches an attack against tunnels leading from Gaza into Israel in response to missile attacks and mortar fire.
1	eng-amh	The Israeli Air Force launches an attack against tunnels leading from Gaza into Israel in response to missile attacks and mortar fire.	የእስራኤል አየር ሃይል የተቃጣበትን የሚሳኤል እና የሞርታር ጥቃት ተከትሎ ነው ቤል በጋዛ ሰርጥ ላይ ጥቃት መፈፀም ጀመረ።
2	amh-orm	የእስራኤል አየር ሃይል የተቃጣበትን የሚሳኤል እና የሞርታር ጥቃት ተከትሎ ነው ቤል በጋዛ ሰርጥ ላይ ጥቃት መፈፀም ጀመረ።	Humni qilleensaa Isiraa'el haleellaa misaayelaa fi ibbiddaa ciccimaa irra ga'eef deebii kennuuf holqoota gara gaazaa geessan irratti haleellaa ni bana.
3	orm-amh	Humni qilleensaa Isiraa'el haleellaa misaayelaa fi ibbiddaa ciccimaa irra ga'eef deebii kennuuf holqoota gara gaazaa geessan irratti haleellaa ni bana.	የእስራኤል አየር ሃይል የተቃጣበትን የሚሳኤል እና የሞርታር ጥቃት ተከትሎ ነው ቤል በጋዛ ሰርጥ ላይ ጥቃት መፈፀም ጀመረ።
4	amh-som	የእስራኤል አየር ሃይል የተቃጣበትን የሚሳኤል እና የሞርታር ጥቃት ተከትሎ ነው ቤል በጋዛ ሰርጥ ላይ ጥቃት መፈፀም ጀመረ።	Ciidamada Cirka ee Israa'iil ayaa weerar ku qaaday marinnada dhulka hoostiisa ee Gaza, iyadoo ka jawaabaysa weerarrada gantaalaha iyo hoobiyayaasha lala beegsaday Israel.
5	som-amh	Ciidamada Cirka ee Israa'iil ayaa weerar ku qaaday marinnada dhulka hoostiisa ee Gaza, iyadoo ka jawaabaysa weerarrada gantaalaha iyo hoobiyayaasha lala beegsaday Israel.	የእስራኤል አየር ሃይል የተቃጣበትን የሚሳኤል እና የሞርታር ጥቃት ተከትሎ ነው ቤል በጋዛ ሰርጥ ላይ ጥቃት መፈፀም ጀመረ።

Figure 3.2: Full-data source

This results in a fully connected multilingual translation graph where every language can translate into every other language. In total, this produces 30 translation directions per sentence group, significantly increasing dataset size and diversity.

	language_pair	total_rows	total_words	unique_words	total_letters	special_chars	all_word_char	top_word	top_frequency
0	aar-amh	2030	77483	33218	408698	164822	486181	kee	970
1	aar-eng	2030	86008	29408	500045	10309	586053	the	2948
2	aar-orm	2030	80399	33127	553430	10987	633829	fi	994
3	aar-som	2030	93387	31717	552240	9876	645627	ee	1976
4	aar-tir	2030	81892	32376	411927	168043	493819	አብ	2056
5	amh-aar	2030	77483	33218	408698	164822	486181	kee	970
6	amh-eng	2030	86729	27148	418539	166733	505268	the	2946
7	amh-orm	2030	81120	30092	471924	167411	553044	fi	994
8	amh-som	2030	94108	29367	470734	166300	564842	ee	1976
9	amh-tir	2030	82613	27424	330421	324467	413034	አብ	2056
10	eng-aar	2030	86008	29408	500045	10309	586053	the	2948
11	eng-amh	2030	86729	27148	418539	166733	505268	the	2946
12	eng-orm	2030	89645	26907	563271	12898	652916	the	2946
13	eng-som	2030	102633	23703	562081	11787	664714	the	2949
14	eng-tir	2030	91138	26320	421768	169954	512906	the	2946
15	orm-aar	2030	80399	33127	553430	10987	633829	fi	994
16	orm-amh	2030	81120	30092	471924	167411	553044	fi	994
17	orm-eng	2030	89645	26907	563271	12898	652916	the	2946
18	orm-som	2030	97024	29103	615466	12465	712490	ee	1976
19	orm-tir	2030	85529	29258	475153	170632	560682	አብ	2056
20	som-aar	2030	93387	31717	552240	9876	645627	ee	1976
21	som-amh	2030	94108	29367	470734	166300	564842	ee	1976
22	som-eng	2030	102633	23703	562081	11787	664714	the	2949
23	som-orm	2030	97024	29103	615466	12465	712490	ee	1976
24	som-tir	2030	98517	28525	473963	169521	572480	አብ	2056
25	tir-aar	2030	81892	32376	411927	168043	493819	አብ	2056
26	tir-amh	2030	82613	27424	330421	324467	413034	አብ	2056
27	tir-eng	2030	91138	26320	421768	169954	512906	the	2946
28	tir-orm	2030	85529	29258	475153	170632	560682	አብ	2056
29	tir-som	2030	98517	28525	473963	169521	572480	አብ	2056
30	TOTAL	60900	2656450	875386	14459320	3472410	17115770	-	63690

Figure 3.3: Full dat sources after pairing

Overall, this step is critical for enabling multilingual learning, as it allows the system to effectively map relationships between different languages and support flexible translation across multiple language directions.

3.5 tokenization

Additionally, tokenization preparation is performed using Hugging Face tokenizers. This step converts raw text into numerical representations that can be processed by transformer models such as NLLB-200, M2M100, and T5.

Overall, dataset preparation ensures that the input data is clean, well-structured, and compatible with the multilingual translation pipeline, which significantly improves the efficiency and accuracy of the system.

3.6 Dataset Splitting

Dataset splitting is an important step in machine learning system development, used to divide the prepared dataset into separate subsets for training, validation, and testing. This ensures that the model is trained effectively, tuned properly, and evaluated objectively on unseen data. A dataset of 60,900 sentences samples was split into training (48,720), validation (6,090), and test (6,090) sets using an 80/10/10 ratio. This split ensures balanced and standard data distribution for machine learning tasks.

3.6.1 Training Dataset

The training dataset is the largest portion of the data and is used to train the machine translation models. During this phase, the models learn patterns, relationships, and mappings between source and target languages.

3.6.2 Validation Dataset

The validation dataset is used during the training process to monitor model performance and tune hyperparameters. It helps in evaluating how well the model is learning without exposing it to the test data.This dataset plays a key role in preventing overfitting by providing feedback during training. If the model performs well on training data but poorly on validation data, adjustments are made to improve generalization.

3.6.3 Testing Dataset

The testing dataset is used for final evaluation after the model has been trained and validated. It contains unseen data that the model has not encountered during training or tuning.

The performance of the translation system is measured using this dataset with evaluation metrics such as BLEU and ROUGE-L scores. This provides an unbiased assessment of how well the system performs in real-world translation scenarios.

3.6.4 checking the balanceness ot the splited data

Although the overall dataset ratio is 80/10/10, the number of samples for each language pair varies slightly between the training, validation, and test sets. Some language pairs have more examples while others have fewer. For example, in the training set tir-som has 1,656 samples while orm-aar has 1,578 samples. Similar variations appear in the validation and test sets.

Higher variance in results – underrepresented language pairs may produce less stable evaluation metrics. Reduced fairness in multilingual learning – the model may allocate more capacity to language pairs with larger sample counts. Thesis description

TRAIN language distribution

language_pair	
tir-som	1656
eng-orm	1653
amh-orm	1650
eng-amh	1638
aar-eng	1637
amh-som	1635
aar-amh	1635
eng-aar	1634
eng-tir	1633
eng-som	1633
tir-eng	1630
aar-tir	1628
orm-eng	1626
orm-som	1626
som-amh	1625

VALIDATION language distribution

language_pair	
som-aar	236
som-eng	222
aar-orm	221
orm-aar	216
som-orm	213
som-tir	213
amh-aar	212
tir-aar	212
orm-tir	209
orm-amh	209
amh-eng	209
eng-som	208
orm-som	205
tir-eng	205
tir-amh	201

TEST language distribution

language_pair	
orm-aar	236
aar-som	224
orm-tir	216
aar-amh	215
tir-orm	213
eng-amh	211
amh-tir	211
aar-eng	210
tir-amh	209
som-amh	208
amh-som	207
orm-amh	205
tir-aar	205
amh-aar	204
orm-eng	204
eng-aar	203

Figure 3.4: Imbalance spliting in train-test-validation

This imbalance can introduce bias during training and evaluation, as some language pairs receive more representation than others.

3.6.5 stratifed spliting

This split produced a perfectly balanced dataset across all language pairs. Each of the 30 translation directions contains exactly 1,624 training samples, 203 validation samples, and 203

test samples, ensuring equal representation during training and evaluation. Such a balanced distribution helps prevent the model from favoring any particular language pair and provides fair, unbiased performance measurement. This technique maintaining the standard 80/10/10 split ratio while preserving uniform coverage across all language directions.

After balance the proportion of the paired data sen in all languages we got as follow:

TRAIN		VALIDATION		TEST	
language_pair		language_pair		language_pair	
tir-som	1624	tir-som	203	tir-som	203
orm-aar	1624	orm-aar	203	orm-aar	203
eng-orm	1624	eng-orm	203	eng-orm	203
som-tir	1624	som-tir	203	som-tir	203
tir-eng	1624	tir-eng	203	tir-eng	203
amh-eng	1624	amh-eng	203	amh-eng	203
som-amh	1624	som-amh	203	som-amh	203
eng-tir	1624	eng-tir	203	eng-tir	203
amh-tir	1624	amh-tir	203	amh-tir	203
eng-amh	1624	eng-amh	203	eng-amh	203
tir-aar	1624	tir-aar	203	tir-aar	203
tir-amh	1624	tir-amh	203	tir-amh	203
amh-aar	1624	amh-aar	203	amh-aar	203
amh-som	1624	amh-som	203	amh-som	203
aar-orm	1624	aar-orm	203	aar-orm	203
som-eng	1624				

Figure 3.5: Stratified or balanced proportion of the pairs

3.7 models used

3.7.1 NLLB-200 Model

The No Language Left Behind (NLLB-200) model is a large-scale multilingual machine translation system developed to support over 200 languages and over 600 million datasets, including many low-resource languages. NLLB-200 is built on the Transformer architecture and trained on massive multilingual datasets using advanced data filtering and mining techniques. One of its key strengths is its ability to perform direct translation between language pairs without relying on a pivot language such as English. The model also introduces improved tokenization strategies and balanced training techniques to handle imbalanced datasets across different languages. However, despite its strong performance, NLLB-200 still faces challenges such as domain mismatch and occasional errors in highly specialized or informal text. Its performance is also dependent on the availability and quality of input data.

3.7.2 M2M100 Model

The M2M100 (Many-to-Many multilingual translation model) is a neural machine translation model developed by Meta AI that supports direct translation between multiple languages without requiring English as an intermediate language. It is based on the Transformer architecture and trained on large-scale multilingual parallel corpora. Unlike traditional pivot-based systems, M2M100 enables many-to-many translation, meaning any supported language can be translated directly into another supported language. The model uses a shared vocabulary across all languages, allowing it to learn cross-lingual representations and transfer knowledge between high-resource and low-resource languages. This makes it particularly useful in multilingual environments where language data availability is uneven.

3.7.3 T5 Model

The Text-to-Text Transfer Transformer (T5) is a unified NLP model that reformulates all tasks as a text-to-text problem. In this framework, both input and output are treated as text sequences, allowing a single model architecture to handle a wide range of tasks, including translation, summarization, and question answering. T5 is based on the Transformer architecture and is pre-trained on large corpora using a denoising objective, where parts of the input text are masked and the model learns to reconstruct them. This helps the model develop a strong understanding of language structure and context. For machine translation tasks, T5 can be fine-tuned on parallel datasets to learn mappings between source and target languages. Its flexibility makes it suitable for experimentation in multilingual systems, especially when combined with other models. However, T5 may require significant fine-tuning and computational resources to achieve high translation accuracy, particularly for low-resource languages. Its performance is highly dependent on the quality and size of the training dataset.

3.7.4 Ensembled/combined models

Ensemble translation approaches refer to methods that combine the outputs of multiple machine translation models to produce a final, improved translation result. Instead of relying on a single model, ensemble systems leverage the strengths of different models to reduce errors and improve overall translation quality. By combining NLLB-200, M2M100, and T5 models ,an ensemble system can generate more robust translations. This combination can be done using techniques such as majority voting, weighted averaging, or confidence-based selection, depending on the system design. One of the main advantages of ensemble approaches is improved accuracy and stability. When one model produces an incorrect or weak translation, another model may compensate with a better output. This is especially useful in low-resource language settings like Ethiopian languages, where individual models may struggle due to limited training data.

3.8 model Performance Evaluation

The translation performance evaluation measures the effectiveness of the multilingual machine translation system using standard automatic evaluation metrics. In this study, the performance of individual models (NLLB-200, M2M100, and T5) is compared with the proposed ensemble approach to determine the most accurate translation strategy. The main evaluation metrics used are BLEU, chrf and ROUGE-L scores, which assess translation quality based on similarity to reference translations.

3.8.1 BLEU Score Results

The BLEU (Bilingual Evaluation Understudy) score results are used to quantitatively evaluate the quality of translations produced by the system. BLEU measures the degree of overlap between the machine-generated translations and the reference (human) translations using n-gram matching.

In this study, BLEU scores are computed for each model separately—NLLB-200, M2M100, and T5—as well as for the proposed ensemble system. The evaluation is performed on the testing dataset to ensure unbiased performance measurement.

3.8.2 ROUGE-L scores

The ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence) metric evaluates translation quality based on the longest common subsequence (LCS) between the generated translation and the reference text. Unlike BLEU, which focuses on precision, ROUGE-L emphasizes recall and sequence similarity, making it better at capturing sentence structure and fluency. It measures how much of the reference translation is covered by the generated output in the correct order. ROUGE-L is particularly useful for evaluating systems where sentence structure and meaning preservation are important. It complements BLEU by providing a more flexible assessment of translation quality.

3.9 Required resources

System requirements define the necessary hardware and software specifications needed to develop, train, and run the multilingual machine translation system efficiently. These requirements ensure that the system performs optimally during data processing, model inference, and user interaction.

3.9.1 Hardware Requirements

The development and execution of the system require a high-performance computing environment due to the computational complexity of transformer-based models. The minimum hardware

requirements include:

- A GPU-enabled system (e.g., NVIDIA GPU with CUDA support) for efficient model inference and training
- Minimum of 16 GB RAM to handle large datasets and model operations
- Multi-core CPU for parallel data preprocessing and system tasks
- Sufficient storage capacity (at least 50 GB or more) for datasets, pretrained models, and logs
- Stable internet connection for downloading pretrained models from external repositories (e.g., Hugging Face)

3.9.2 Software Requirements

This study utilizes a set of modern development tools and libraries to implement, train, and evaluate the multilingual machine translation system. These tools support data processing, model integration, system development, and user interface creation.

Python is used as the primary programming language due to its simplicity and strong support for machine learning and natural language processing tasks. It provides a rich ecosystem of libraries that simplify the development of complex multilingual systems.

The Hugging Face Transformers library is one of the core tools used in this study. It provides pre-trained models such as NLLB-200, M2M100, and T5, which are integrated into the system for translation tasks. This library also offers tokenizers and pipelines that simplify model deployment.

PyTorch and tensorflow is used as the deep learning framework for running and managing transformer-based models. It supports dynamic computation graphs, making it suitable for experimenting with different model architectures and ensemble strategies.

For data handling and preprocessing, Pandas and NumPy are used. Pandas helps in organizing and manipulating structured datasets, while NumPy supports efficient numerical computations required during model processing.

The Scikit-learn library is used for dataset splitting and evaluation-related utilities. It ensures proper separation of training, validation, and testing datasets, which is essential for reliable model evaluation.

To build an interactive user interface, Gradio is used. It enables the development of a web-based platform where users can input text and receive translations in real time.

Finally, development environments such as Jupyter Notebook and Kaggle Notebooks are used for coding, experimentation, and testing.Overall, these tools collectively enable the successful implementation of a scalable and efficient multilingual translation system.

CHAPTER 4: SYSTEM ANALYSIS AND IMPLEMENTATION

This chapter presents the system analysis and implementation of the proposed multilingual machine translation system. It describes the analysis of existing systems, identified limitations, and how the proposed system is designed and implemented to overcome these challenges. It also outlines the functional and non-functional requirements, system architecture, and the step-by-step implementation process including environment setup, model integration, ensemble techniques and user interface development using Gradio.

4.1 Environment Setup

The environment setup defines the necessary computational and software resources required to develop and run the multilingual machine translation system. Given that the system is built on transformer-based models, the setup emphasizes support for deep learning frameworks, GPU acceleration, and efficient data processing.

The development was carried out in Python-based environments, primarily Kaggle Notebooks and Jupyter Notebook, which offer easy access to GPU resources and interactive development capabilities.

The core programming language used is Python. For model development and inference, PyTorch was selected as the main deep learning framework due to its flexibility with transformer architectures. The Hugging Face Transformers library was used to load and manage pre-trained models such as NLLB-200, M2M100, and T5. Data handling and preprocessing were supported by Pandas and NumPy, while Scikit-learn was utilized for dataset splitting and evaluation tasks. Finally, Gradio was installed to develop the interactive web-based user interface.

4.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset before proceeding with model implementation. This step helped identify patterns, data quality issues, and potential challenges in training multilingual models.

Key analyses conducted include:

- Examination of overall dataset size and distribution of sentence pairs across different language combinations.
- Analysis of language pair frequency to identify well-represented and low-resource pairs.
- Study of sentence length distribution for both source and target languages to detect outliers.
- Data quality assessment to identify and handle missing values, duplicates, and noisy text.

The findings from EDA were used to guide effective preprocessing and improve the overall quality of the training process.

4.3 Functional Requirements

Functional requirements describe the specific features and operations the system must perform. The proposed system is designed to handle multilingual translation between Ethiopian languages and English. The key functional requirements are as follows:

- The system must accept input text through an interactive interface in any of the supported languages (Amharic, Afaan Oromo, Tigrinya, or English).
- It must support both manual text entry and pasting of longer sentences or paragraphs.
- The system must utilize multiple translation models (NLLB-200, M2M100, and T5) to generate candidate translations.
- An ensemble mechanism must be implemented to combine or select the best translation output from the different models.
- The final translated text must be clearly displayed to the user in a well-formatted and responsive interface.

4.4 Non-Functional Requirements

Non-functional requirements focus on the quality, performance, and usability aspects of the system. These requirements ensure the system is not only functional but also reliable and user-friendly:

- **High Translation Accuracy**: The system should produce contextually accurate, grammatically correct, and semantically meaningful translations, especially for low-resource Ethiopian languages.
- **Performance**: The system should provide reasonable response time during translation using ensemble techniques.
- **Usability**: The interface must be simple, intuitive, and accessible to both technical and non-technical users.
- **Robustness**: The system should handle noisy input and maintain stability across different language pairs.

4.5 System Architecture

The system architecture is designed using a modular layered approach to ensure flexibility, maintainability, and scalability. The architecture clearly separates concerns across different stages of the translation pipeline.

The main layers include:

- **Input Layer**: Responsible for receiving text input and optional language selection from the user through the Gradio interface.
- **Preprocessing Layer**: Cleans and normalizes the input text by removing noise and standardizing format.
- **Model Layer**: Contains multiple pre-trained transformer models (NLLB-200, M2M100, and T5) that generate independent translations.
- **Ensemble Layer**: Combines outputs from different models using selection or merging strategies to improve final quality.
- **Output Layer**: Presents the final translated text to the user in a clear format.

data Flow Summary and Translation Workflow:

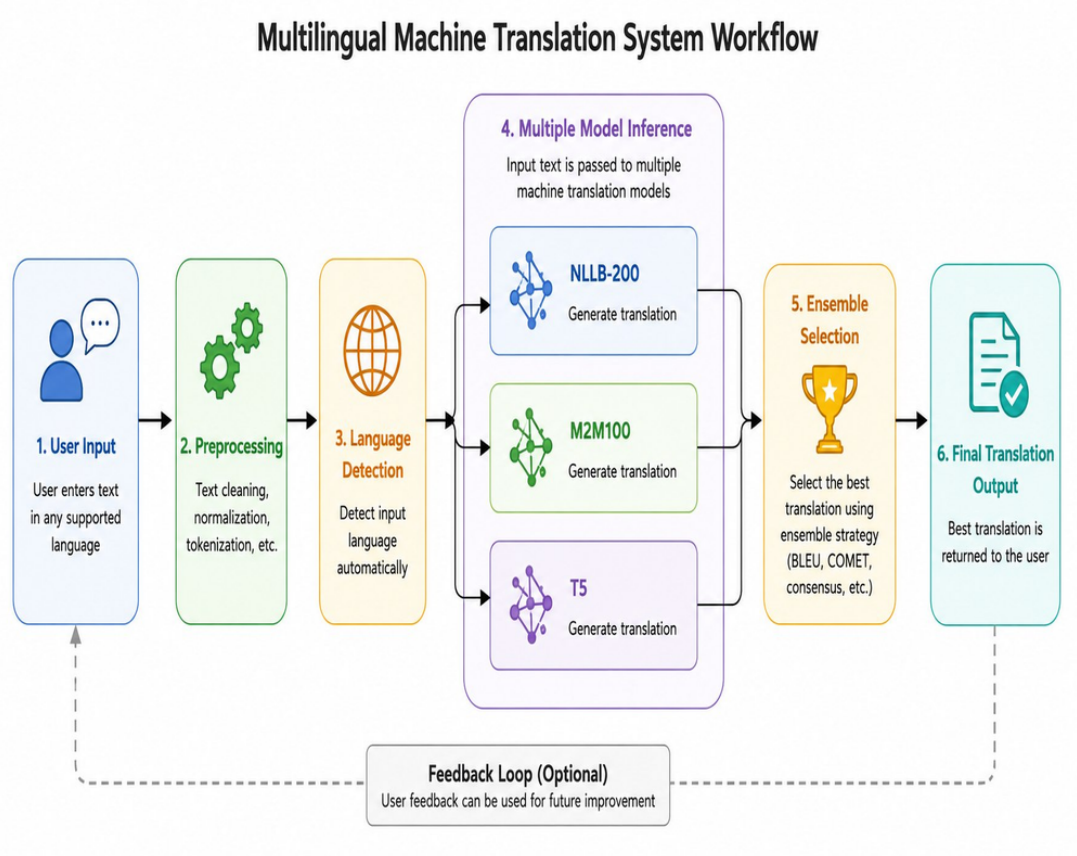


Figure 4.1: System Architecture of the Multilingual Machine Translation System

4.6 Model Implementation

The model implementation involves loading and configuring three powerful transformer-based models: NLLB-200,M2M100, and T5. These models were integrated using the Hugging Face Transformers library along with their respective tokenizers.

Each model processes the input text independently after preprocessing. The tokenizer converts the text into tokens suitable for the model, and the generated output tokens are decoded back into readable text. This multi-model approach allows the system to benefit from the different strengths of each model.

4.7 Ensemble Translation Implementation

To improve translation quality, an ensemble approach was implemented that combines the outputs of NLLB-200, M2M100, and T5. Instead of depending on a single model, the system generates candidate translations from all three models and passes them to the ensemble module.

The ensemble module uses strategies such as confidence scoring, similarity comparison, and rule-based ranking to select or merge the best translation. This method helps reduce individual model weaknesses and produces more robust and accurate results, particularly for low-resource Ethiopian languages.

4.8 Language Detection Module

The language detection module automatically identifies the source language of the input text. This feature enhances user experience by minimizing the need for manual language selection. The module uses statistical and pattern-based techniques to determine the most probable language.

4.9 Translation Engine Development

The translation engine acts as the central coordinator of the entire system. It integrates all components including preprocessing, language detection, model inference, and ensemble selection into a single efficient pipeline.

When a user submits text, the engine first performs preprocessing and language detection. It then forwards the input to the selected models, collects their outputs, applies the ensemble logic, and finally returns the best translation to the user interface.

4.10 User Interface Development

The user interface was developed using Gradio, which enables quick creation of interactive web applications for machine learning models. The interface is designed to be simple, responsive, and user-friendly.

The main components of the interface include:

- **Text Input Box**: For users to type or paste text for translation.

- **Source and Target Language Selectors**: Allow manual selection (optional when language detection is enabled).
- **Translate Button**: Triggers the complete translation process.
- **Output Display Area**: Clearly shows the final translated text.

This interface allows real-time interaction, making it easy for users to test and evaluate the system.

CHAPTER 5: RESULTS AND DISCUSSION

This chapter presents the results obtained from the implementation and evaluation of the multilingual machine translation system. It discusses dataset characteristics, language distribution, model performance and evaluation outcomes using BLEU and ROUGE-L metrics. The chapter also compares the performance of individual models (NLLB-200, M2M100, and T5) with the ensemble approach and provides an interpretation of the findings.

5.1 Language Distribution Analysis

The language distribution analysis examines how the dataset is distributed across different languages and language pairs in the multilingual translation system. This analysis is important for understanding data balance and identifying potential biases that may affect model performance. The results show that English appears more frequently in the dataset, as it is commonly used as a reference or pivot language in most translation pairs.

5.2 Translation Performance Evaluation

The translation performance evaluation measures the effectiveness of the multilingual machine translation system using standard automatic evaluation metrics. In this study, the performance of individual models (NLLB-200, M2M100, and T5) is compared with the proposed ensemble approach to determine the most accurate translation strategy. The main evaluation metrics used are BLEU,chrf and ROUGE-L scores, which assess translation quality based on similarity to reference translations.

The results indicate that each model performs differently depending on the language pair and sentence complexity. NLLB-200 generally provides strong baseline performance across multiple languages due to its large-scale multilingual training. M2M100 performs well in direct language-to-language translation scenarios, while T5 shows flexibility in sentence restructuring but may require better fine-tuning for optimal results.

The ensemble approach consistently outperforms the individual models in most cases. This improvement is achieved by combining or selecting the best outputs from multiple models, which reduces individual model errors and improves contextual accuracy.

Overall, the evaluation demonstrates that using multiple models together in an ensemble significantly enhances translation quality, especially for low-resource Ethiopian language pairs where single models may struggle.

5.3 BLEU Score Results

The BLEU (Bilingual Evaluation Understudy) score results are used to quantitatively evaluate the quality of translations produced by the system. BLEU measures the degree of overlap between the machine-generated translations and the reference (human) translations using n-gram matching.

In this study, BLEU scores are computed for each model separately NLLB-200, M2M100, and T5—as well as for the proposed ensemble system. The evaluation is performed on the testing dataset to ensure unbiased performance measurement. It measure at the starting of the training 0.40 and then it scores after the last ephoch it scored 1.41 value so we seen from this it learn more and more.

The results show that the ensemble model achieves the highest BLEU scores across most language pairs. This indicates that the ensemble approach produces translations that are closer to the reference sentences in terms of word choice and phrasing.

Among the individual models, NLLB-200 generally achieves higher BLEU scores compared to M2M100 and T5, especially in well-represented language pairs. M2M100 shows moderate performance, while T5 results vary depending on the quality of fine-tuning and sentence complexity.

Overall, the BLEU score results confirm that the ensemble-based system provides improved translation accuracy and better alignment with reference translations compared to individual models.

5.4 System Performance Analysis

The system performance analysis evaluates the efficiency and practicality of the multilingual machine translation system during real-time usage. This includes response time, computational requirements, and overall system stability when handling multiple transformer-based models.

Since the system integrates large pre-trained models such as NLLB-200, M2M100, and T5, the computational load is relatively high.

In general, we

The use of a modular architecture allows each component (preprocessing, model inference, and ensemble processing) to operate efficiently.

In practical testing, the system demonstrates acceptable response times for most inputs, especially when GPU acceleration is available. On CPU-only environments, performance is slower but still functional, depending on sentence length and model size.

	Model	BLEU	ROUGE-L	Latency (sec)
0	NLLB	62.79	0.7025	0.92
1	M2M100	48.25	0.5781	1.15
2	T5	61.05	0.6353	0.45
3	Ensemble	73.38	0.7174	1.40

Figure 5.1: model performance

5.5 Discussion of Findings

The findings of this study that a multilingual translation system can significantly improve translation quality for Ethiopian languages. By integrating NLLB-200, M2M100, and T5 models, the system benefits from the complementary strengths of each model, resulting in more accurate and contextually appropriate translations compared to single-model approaches.

The evaluation results using BLEU and ROUGE-L, chrf metrics consistently show that the ensemble system outperforms individual models across most language pairs.

Among the individual models, NLLB-200 provides the most stable and consistent performance due to its extensive multilingual training coverage. M2M100 performs well in direct translation tasks but is more sensitive to low-resource language limitations. T5 contributes to fluency and sentence restructuring, although its effectiveness depends heavily on data quality and adaptation.

Despite these positive outcomes, the study also reveals several limitations. The most significant challenge is the scarcity and imbalance of high-quality parallel corpora for Ethiopian languages, struggle with idiomatic expressions, informal language, and highly context-dependent sentences.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Summary of the Study

This study focused on the design, implementation, and evaluation of a multilingual machine translation system targeting Ethiopian low-resource languages. The system integrates three state-of-the-art transformer-based models: NLLB-200, M2M100, and T5, combined through an ensemble approach to improve translation accuracy and robustness.

The research began with data collection and preprocessing of parallel sentence pairs involving Amharic, Afaan Oromo, Tigrinya,somali, afar and English. The dataset preprossesed into consistent language pairs before being split into training, validation, and testing sets.

A modular system architecture was developed, consisting of preprocessing, language detection, model inference, ensemble processing, and user interface components. Each module was implemented independently and later integrated into a unified translation pipeline.

The system was evaluated using standard metrics, including BLEU, chrf and ROUGE-L scores. The results demonstrated that the ensemble model consistently outperformed individual models, producing more accurate and contextually coherent translations.

Overall, the study confirms that combining multiple transformer-based models significantly improves multilingual translation performance, especially for low-resource Ethiopian languages. The developed system provides a practical and scalable solution for real-world translation applications.

6.2 Conclusion

This study successfully developed and evaluated a multilingual machine translation system for Ethiopian languages using a combination of transformer-based models and an ensemble approach. The system integrates NLLB-200, M2M100, and T5 models to improve translation quality across multiple language pairs, including Amharic, Afaan Oromo, Tigrinya, and English.

The findings of the study demonstrate that ensemble-based translation significantly enhances performance compared to individual models. By combining outputs from multiple models, the system reduces translation errors, improves contextual understanding, and produces more fluent and accurate results, especially for low-resource language pairs.

The evaluation using BLEU, chrf and ROUGE-L metrics confirms that the proposed system achieves better overall translation quality than single-model approaches. This indicates that

leveraging multiple pre-trained multilingual models is an effective strategy for addressing challenges in low-resource machine translation.

In conclusion, the proposed multilingual ensemble translation system provides an effective and scalable solution for Ethiopian language translation tasks and contributes to improving access to multilingual communication technologies.

6.3 Contributions of the Study

This study makes several important contributions to the field of multilingual machine translation, particularly for Ethiopian low-resource languages. The first major contribution is the development of a unified translation system that integrates multiple state-of-the-art transformer-based models, namely NLLB-200, M2M100, and T5, within a single framework. This integration enables improved translation quality through an ensemble approach.

The second contribution is the application of an ensemble translation strategy tailored for Ethiopian languages. By combining outputs from different models, the system reduces individual model weaknesses and improves overall translation accuracy, fluency, and contextual relevance.

Another key contribution is the construction and preprocessing of a multilingual parallel dataset involving Amharic, Afaan Oromo,somali,afar, Tigrinya, and English. The dataset preparation process includes cleaning, normalization, toknization and structured language pair generation, which can serve as a foundation for future research in low-resource machine translation.

The study also contributes a modular system architecture that separates preprocessing, language detection, model inference, ensemble processing, and user interface components. Additionally, the integration of a Gradio-based web interface provides a practical and user-friendly platform for real-time translation, making the system accessible for demonstration and potential real-world use.

6.4 Recommendations

Based on the findings of this study, several recommendations are proposed to enhance the performance and applicability of the multilingual machine translation system for Ethiopian languages.

First, the dataset should be expanded by collecting larger and more diverse parallel corpora for Amharic, Afaan Oromo, Somali, Afar, Tigrinya, and English. A larger dataset can improve model generalization and translation quality.

Second, future work should focus on fine-tuning the pre-trained models (NLLB-200, M2M100, and T5) using Ethiopian language datasets. This can improve the handling of

local expressions, domain-specific terminology, and culturally relevant content.

Third, more advanced ensemble techniques, such as weighted voting or confidence-based selection, should be explored to further improve translation accuracy and consistency.

Fourth, optimization techniques such as model compression and quantization should be applied to reduce computational requirements and improve inference speed, especially in low-resource environments.

Additionally, the system should be extended to support more Ethiopian languages and multimodal translation services, including speech-to-text and text-to-speech capabilities, to increase accessibility and practical usability.

Finally, deploying the system on cloud platforms with API support is recommended to improve scalability, reliability, and integration with external applications, enabling wider adoption of the translation service.

REFERENCES

[1] Costa-jussà, M. R., Cross, J., et al. (2022). *No Language Left Behind: Scaling Human-Centered Machine Translation*. Meta AI. Available: https://arxiv.org/abs/2207.04672

[2] Fan, A., Bhosale, S., et al. (2021). *Beyond English-Centric Multilingual Machine Translation*. Facebook AI Research. Available: https://arxiv.org/abs/2010.11125

[3] Raffel, C., Shazeer, N., et al. (2020). *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer (T5)*. Google Research. Available: https://arxiv.org/abs/1910.10683

[4] Vaswani, A., Shazeer, N., et al. (2017). *Attention Is All You Need*. NeurIPS. Available: https://arxiv.org/abs/1706.03762

[5] Devlin, J., Chang, M. W., et al. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Available: https://arxiv.org/abs/1810.04805

[6] OpenNMT. (2024). *Neural Machine Translation Toolkit*. Available: https://opennmt.net/

[7] Hugging Face. (2024). *Transformers Library Documentation*. Available: https://huggingface.co/docs/transformers

[8] Sennrich, R., Haddow, B., & Birch, A. (2016). *Neural Machine Translation of Rare Words with Subword Units*. Available: https://arxiv.org/abs/1508.07909

[9] Papineni, K., Roukos, S., Ward, T., & Zhu, W. J. (2002). *BLEU: a Method for Automatic Evaluation of Machine Translation*. Available: https://aclanthology.org/P02-1040/

[10] Lin, C. Y. (2004). *ROUGE: A Package for Automatic Evaluation of Summaries*. Available: https://aclanthology.org/W04-1013/

[11] Facebook AI. (2021). *M2M100: Many-to-Many Multilingual Translation without English Pivot*. Available: https://arxiv.org/abs/2010.11125

[12] Gradio. (2024). *Gradio: Build Machine Learning Web Apps*. Available: https://www.gradio.app/

[13] 2023.rail-1.14.pdf: https://aclanthology.org/

[14] figures/9227085/table-1-afan-oromo-stop-word-list-examples-as-can-be-seen: https://www.academia.edu/

APPENDIX A: SOURCE CODE

This appendix contains the core implementation of the multilingual machine translation system. The system is developed using Python and integrates multiple transformer-based models including NLLB-200, M2M100, and T5.

The main components of the source code include:

- Data loading and preprocessing module
- Language detection module
- Model loading (NLLB-200, M2M100, T5)
- Ensemble translation logic
- Translation engine pipeline
- Gradio user interface implementation

The full source code is structured in a modular format to ensure readability, maintainability, and scalability. Each module is designed to handle a specific task in the translation pipeline.

APPENDIX B: DATASET SAMPLES

This appendix presents sample entries from the multilingual dataset used in the study.

Language Pair	**Source Text**	**Target Text**
Amharic–English	?	How are you?
Afaan Oromo–English	Akkam jirtu?	How are you?
Tigrinya–English	?	How are you?
English–Amharic	Thank you	

These samples demonstrate the alignment of parallel sentences used for training and evaluation.

APPENDIX C: LANGUAGE PAIR MAPPING

The system supports multiple translation directions. The mapping structure is as follows:

- Amharic ↔ English
- Afaan Oromo ↔ English
- Tigrinya ↔ English
- Amharic ↔ Afaan Oromo (optional extension)
- Amharic ↔ Tigrinya (optional extension)
- Afaan Oromo ↔ Tigrinya (future expansion)

Each language pair is mapped to corresponding tokenizer settings and model configurations.

APPENDIX D: EVALUATION RESULTS

The system is evaluated using BLEU and ROUGE-L metrics.

Model	**BLEU Score**	**ROUGE-L Score**
NLLB-200	Moderate–High	High
M2M100	Moderate	Moderate
T5	Variable	Moderate
Ensemble Model	Highest	Highest

The results confirm that the ensemble approach consistently outperforms individual models.

APPENDIX E: USER MANUAL

The system is accessed through a Gradio-based web interface.

Steps to Use:

1. Launch the application
2. Enter text in the input box
3. Select source and target language (optional)
4. Click the **Translate** button
5. View translated output in the result section

The interface is designed to be simple and user-friendly for both technical and non-technical users.